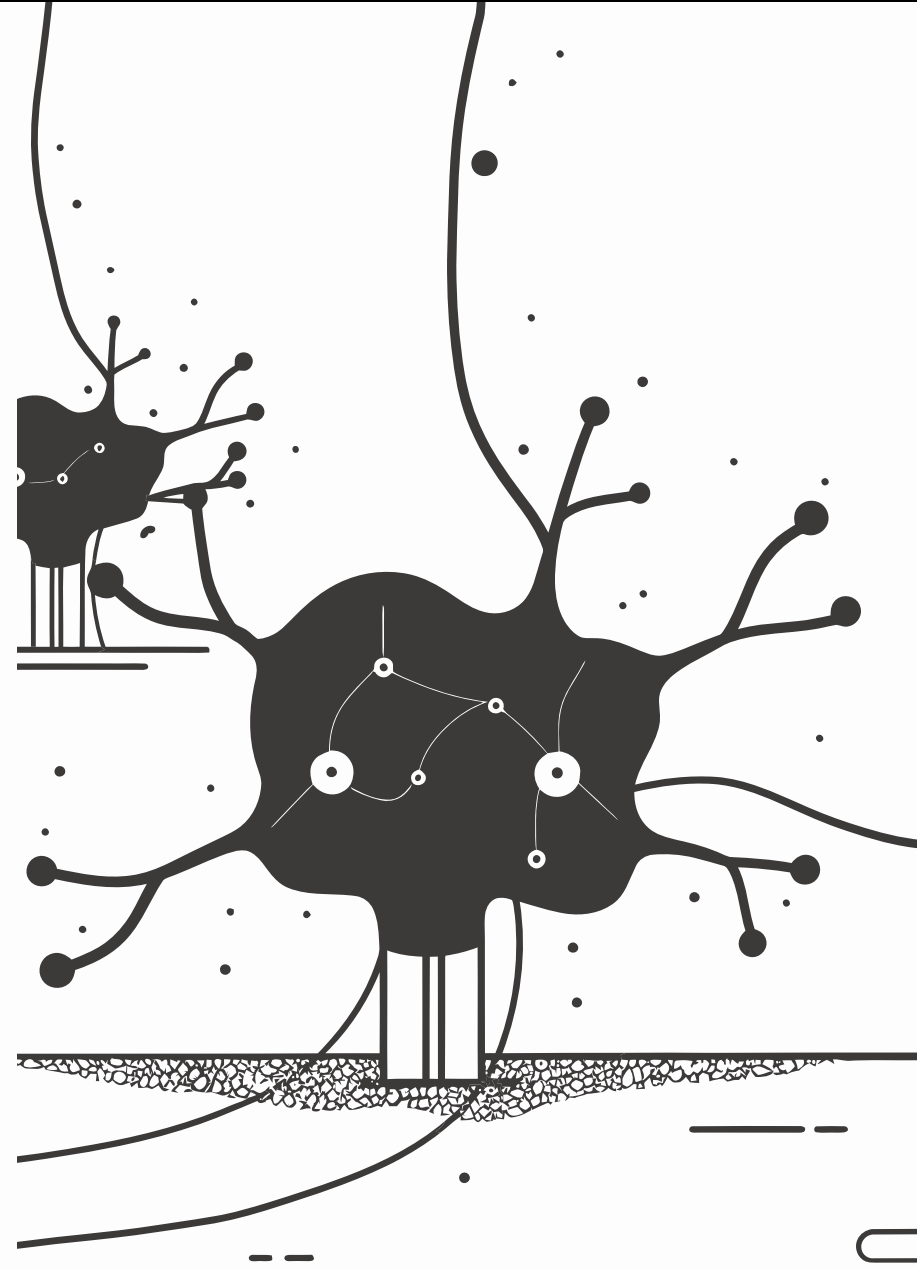


AI Readiness Scale (AIRS)

Development and Validation of a Psychometric Instrument for Assessing Enterprise AI Adoption

DOCTOR OF BUSINESS ADMINISTRATION

N = 523 · OCTOBER–NOVEMBER 2025



The Adoption-Value Gap

88%

AI Tool Usage

Among knowledge workers in 2025, up from
50% in 2023

5%

Value Realized

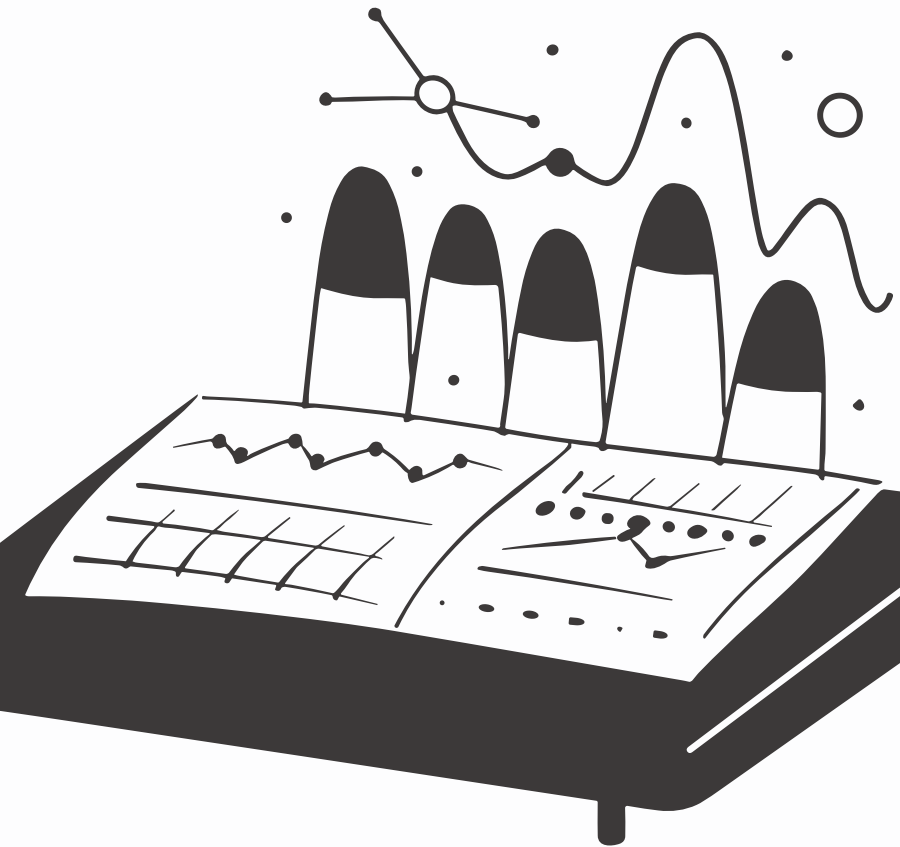
Organizations reporting measurable value from
AI implementations

90–95%

Pilots Stall

Generative AI pilot programs that never
progress beyond initial trials

Organizations invest heavily in AI tools but consistently fail to realize meaningful returns. This adoption-value gap — not a lack of tools, but a failure to convert adoption into value — is the central problem this research addresses.



Three Interconnected Research Gaps

Theoretical Gap

UTAUT2 was validated for consumer mobile technologies. Whether its predictor structure holds for AI – characterized by autonomous decision-making and probabilistic outputs – remained empirically untested.

Measurement Gap

No validated psychometric instrument existed for measuring AI-specific adoption readiness. Existing scales addressed general technology acceptance or narrow AI applications like healthcare robotics.

Practice Gap

Organizations lacked evidence-based frameworks to identify specific barriers and design targeted interventions, leaving AI implementation strategies generic rather than responsive to each workforce's readiness profile.

Research Solution: The AIRS Instrument

What Was Built

The AI Readiness Scale (AIRS) — a 16-item psychometric instrument extending UTAUT2 with an AI Trust construct.

8 factors × 2 items each, plus 4 Behavioral Intention outcome items.

What It Addresses

- Provides a theoretical account of AI adoption drivers
- Offers a practical diagnostic tool for organizational assessment
- Enables targeted, evidence-based intervention design
- Fills all three identified research gaps simultaneously

Theoretical Foundation: UTAUT2

The study built upon UTAUT2 – the most comprehensive and widely validated model of technology acceptance – which integrates seven constructs predicting Behavioral Intention to use a technology.

PE

Performance Expectancy: Perceived improvement in job performance

EE

Effort Expectancy: Perceived ease of learning and use

SI

Social Influence: Degree to which important others encourage use

FC

Facilitating Conditions: Available resources and support

HM

Hedonic Motivation: Pleasure derived from use

PV

Price Value: Perceived cost-benefit balance

HB

Habit: Extent to which use has become automatic

The 8th Construct: AI Trust

AI tools differ from conventional technologies in their capacity for autonomous reasoning, opaque decision processes, and evolving behaviors.

The study introduced **AI Trust (TR)** as an eighth construct — capturing the degree to which users believe AI systems will provide accurate, reliable information and operate in a trustworthy manner.

This extension was grounded in emerging literature identifying trust as a unique barrier to AI adoption not adequately captured by existing UTAUT2 constructs.



Research Questions & Hypotheses

Primary Research Question

To what extent does UTAUT2, extended with AI Trust, explain Behavioral Intention to adopt AI tools among U.S. students and professionals?

Secondary RQs

- RQ2: Factorial validity and psychometric adequacy
- RQ3: Experience and role moderation of structural paths
- RQ4: Relationship between behavioral intention and actual usage
- RQ5: User typology based on readiness patterns

12 Hypotheses Tested

H1a–H1g

7 core UTAUT2 path hypotheses

H2

AI Trust hypothesis

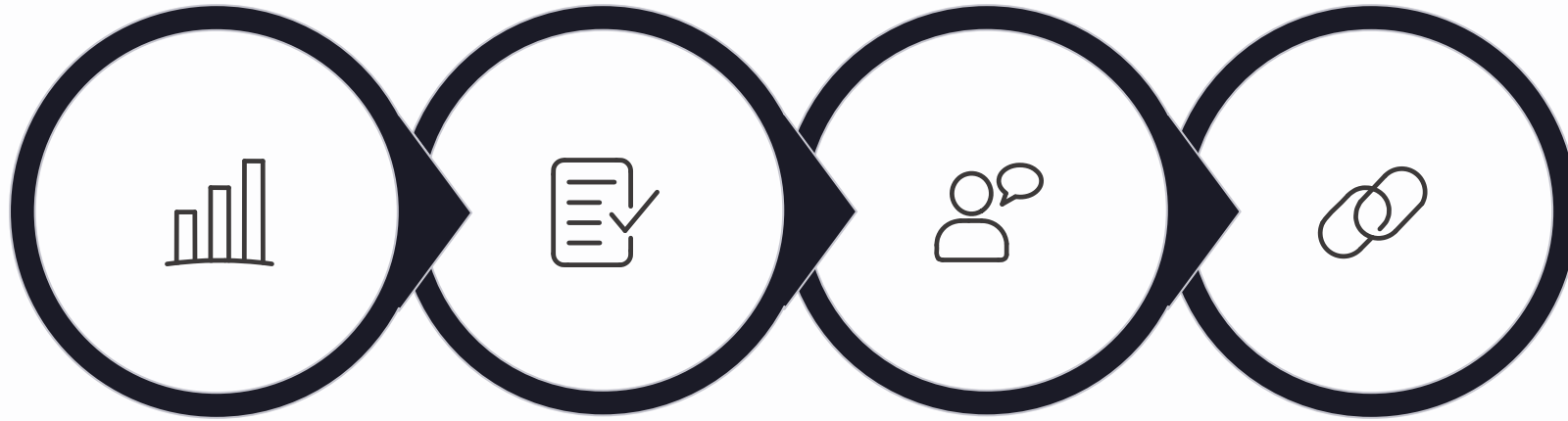
H3–H4

2 moderation hypotheses

H5–H6

2 behavioral validation hypotheses

Methodology: Research Design



Quantitative

Psychometric

Qualitative

Triangulation

A post-positivist, sequential mixed-methods design with quantitative primacy. The quantitative phase provided the primary evidence base; the qualitative phase offered contextual elaboration and triangulation.

Instrument Development Process



Initial Item Pool

28 items across 12 constructs, adapted from validated UTAUT2 scales and supplemented with AI-specific items for Trust, Explainability, Ethical Risk, and AI Anxiety.



Construct Exclusion

Four constructs dropped due to inadequate reliability: Voluntariness ($\alpha = .41$), Explainability ($\alpha = .58$), Ethical Risk ($\alpha = .55$), AI Anxiety ($\alpha = .30$).



Exploratory Factor Analysis

Four competing factor models (A–D) evaluated on the development subsample. Model D – the 8-factor, 16-item solution – selected for optimal structure, reliability, and theoretical interpretability.



Final Instrument

8 factors $\times$ 2 items each = 16 items, plus 4 Behavioral Intention items as the outcome variable.

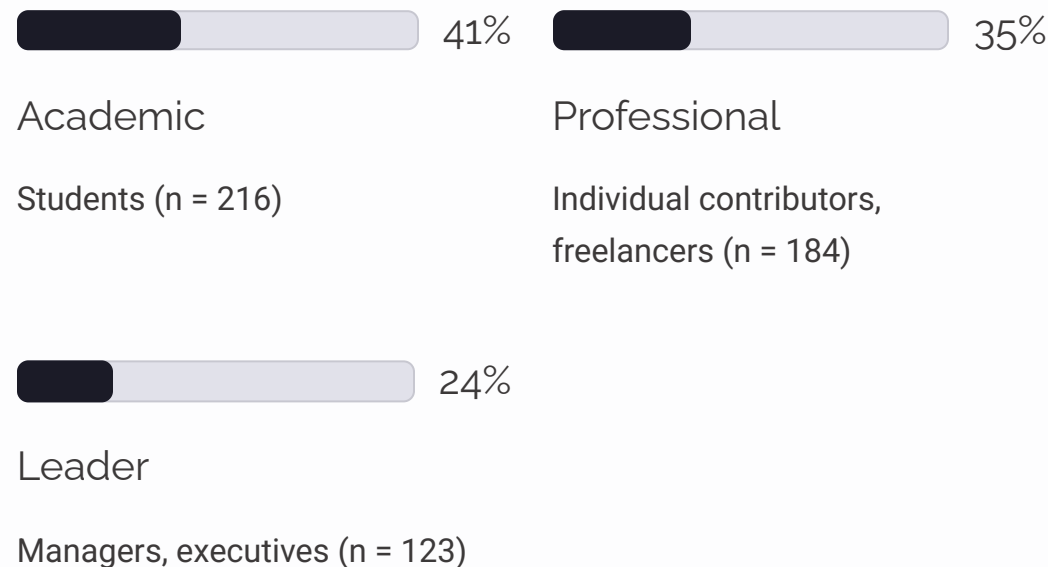
Sample & Data Collection

Sample Overview

N = 523 U.S. adults recruited via Centiment using topic-blinded recruitment — respondents did not know the survey concerned AI, mitigating self-selection bias.

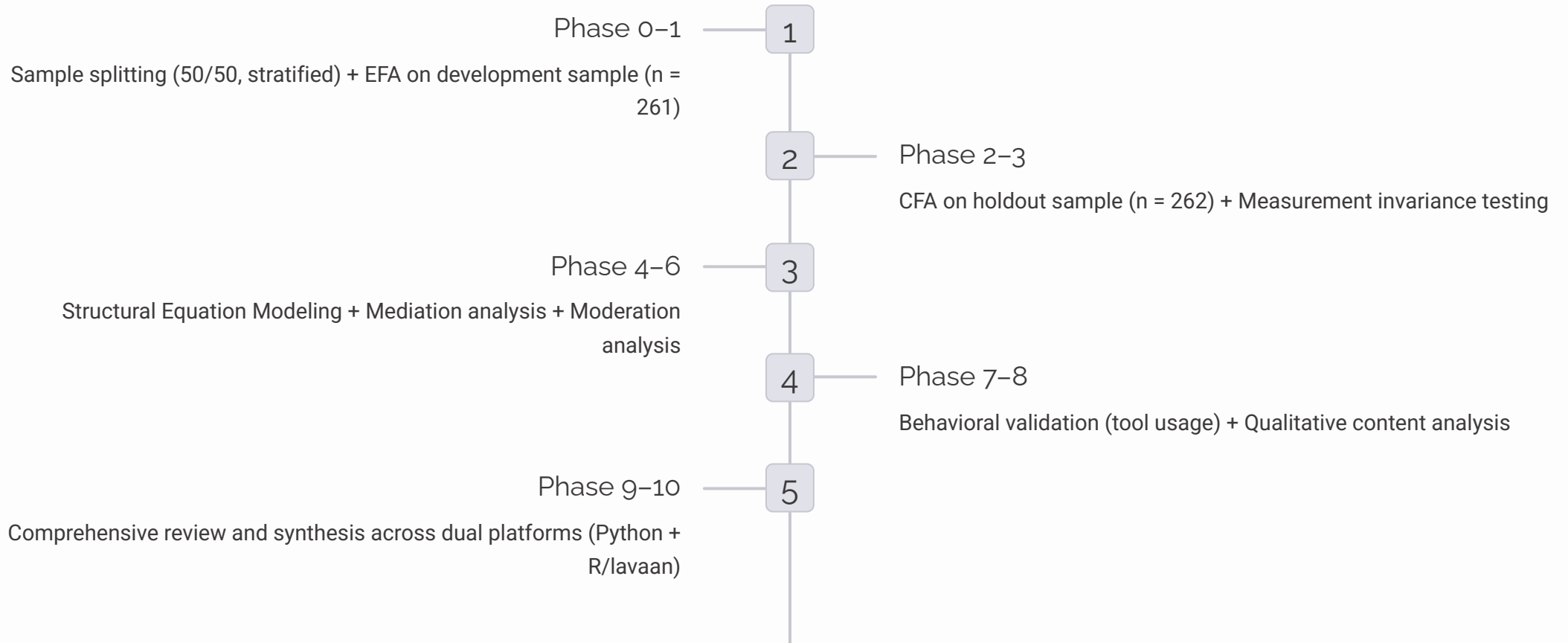
Collection Period: October–November 2025 (3-week window)

Role Distribution



- 68 participants (13.0%) reported a disability — a proportion consistent with U.S. Census Bureau estimates and sufficient for exploratory between-group analysis.

10-Phase Analytical Pipeline



Bootstrap validation used 1,000 resamples for path coefficient robustness. CFA/SEM used Maximum Likelihood with Satorra-Bentler robust corrections (MLM) via R/lavaan 0.6.21.

KEY FINDINGS

Measurement Model: Psychometric Performance

The 16-item data demonstrated excellent suitability for factor analysis: **KMO** = **.937** (superb), Bartlett's Test $\chi^2 = 4,668.45$, $p < .001$, with no multivariate outliers detected.

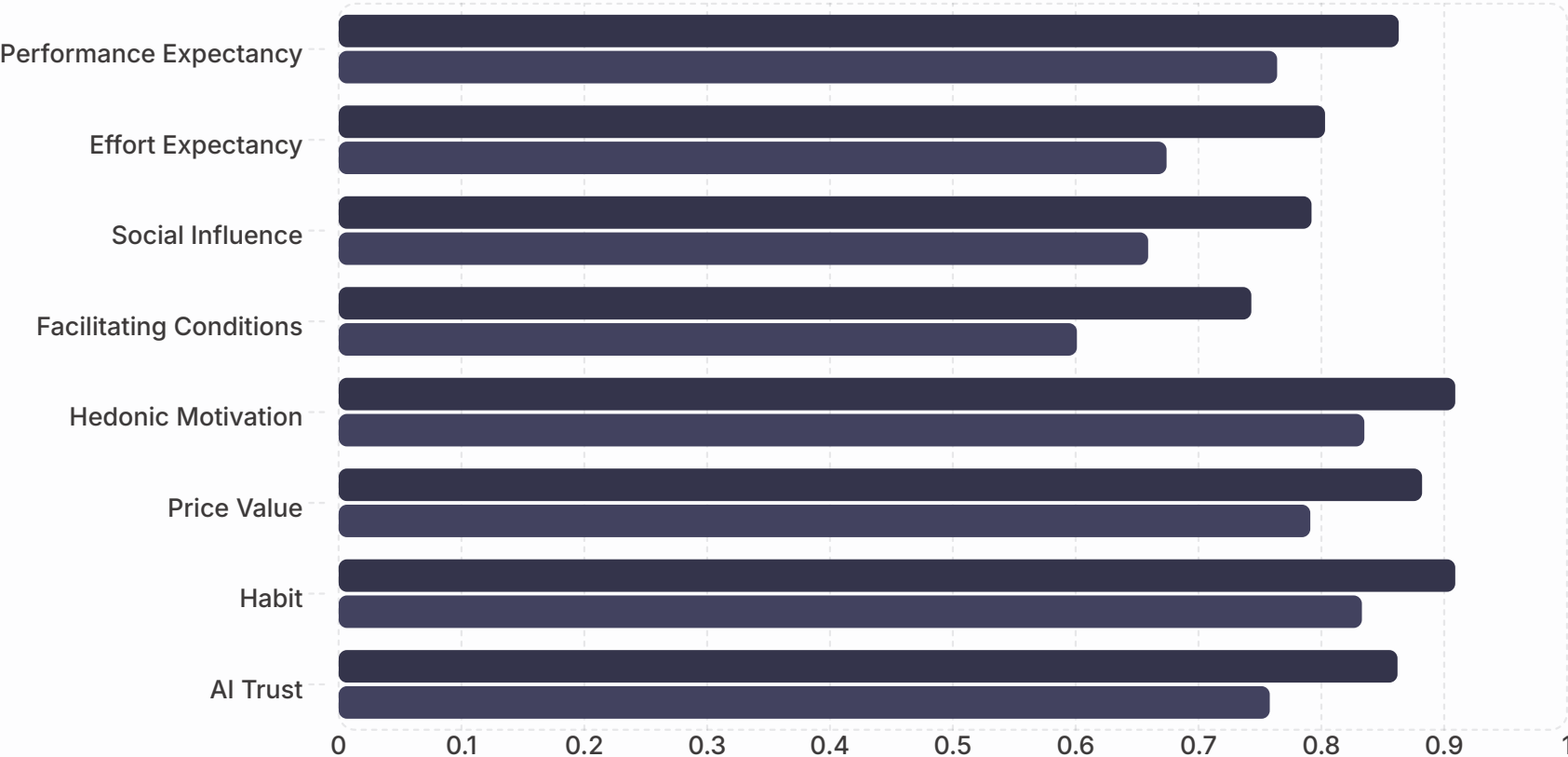
Index	Holdout Value	Full Sample	Threshold	Assessment
CFI	.975	.979	$\geq .95$	Excellent
TLI	.960	.966	$\geq .95$	Excellent
RMSEA	.065	.061	$\leq .08$	Good
SRMR	.026	.022	$\leq .08$	Excellent
χ^2/df	2.10	—	≤ 3.0	Good

Reliability Across All 8 Factors

All eight factors exceeded conventional thresholds for internal consistency. All composite reliabilities exceeded .750 and all AVE values exceeded .50, confirming convergent validity.

■ Cronbach's α ■ AVE

Factor



Discriminant Validity: HTMT Analysis

Factor Correlation Range

r = .516 (EE × SI) to r = .911 (PE × PV)

HTMT Violations (4 pairs)

Pair	HTMT	Status
PE × PV	.902	Exceeds
PE × HM	.900	Exceeds
HM × PV	.904	Exceeds
HM × TR	.850	At threshold

Interpretation

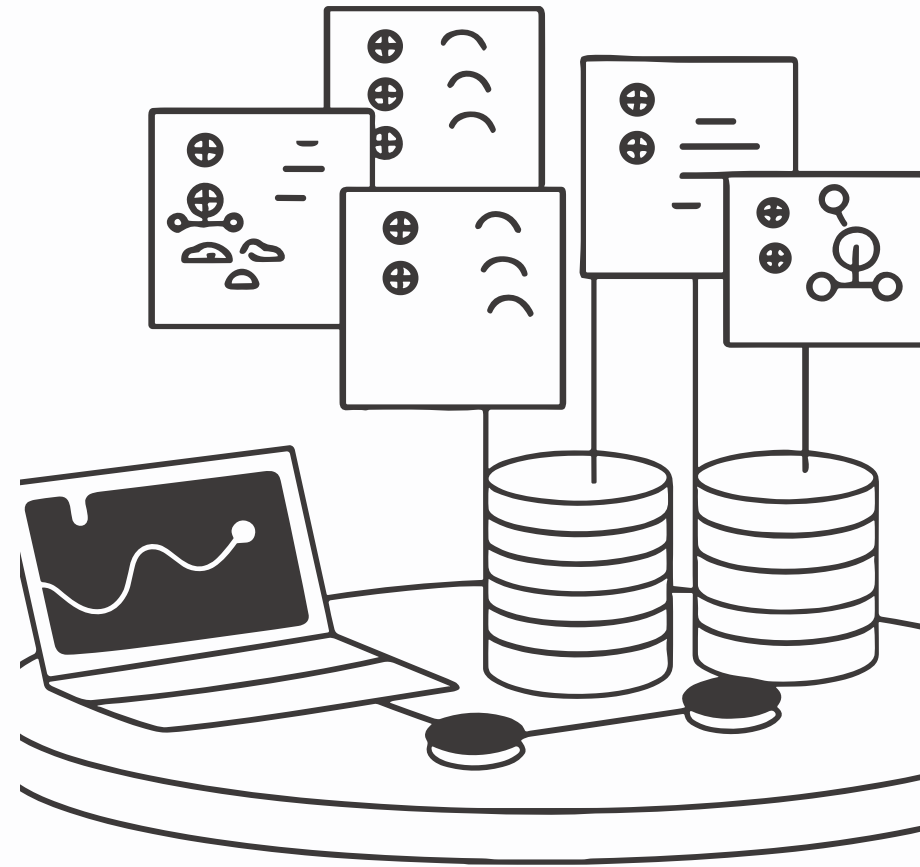
Violations are structurally expected for 2-item-per-factor scales with conceptually adjacent constructs in the PE–HM–PV triad.

The CFA model demonstrated excellent overall fit. The retained 8-factor structure provides essential diagnostic differentiation — each construct measures a **distinct intervention target**.

CENTRAL FINDING

Structural Model: 89.7% Variance Explained

The structural model explained $R^2 = .897$ of the variance in Behavioral Intention — an exceptionally high value indicating that the AIRS factors capture the vast majority of systematic variance in AI adoption intention.



Primary Path Coefficients

H	Path	β	p	Bootstrap 95% CI	Result
H1a	PE → BI	-.028	.791	[-.234, .178]	Not Supported
H1b	EE → BI	-.008	.875	[-.108, .092]	Not Supported
H1c	SI → BI	.136	.024	[.018, .254]	Supported*
H1d	FC → BI	.059	.338	[-.062, .180]	Not Supported
H1e	HM → BI	.217	.014	[.044, .390]	Supported*
H1f	PV → BI	.505	<.001	[.352, .658]	Confirmed – Strongest
H1g	HB → BI	.023	.631	[-.071, .117]	Not Supported
H2	TR → BI	.106	.064	[-.006, .218]	Marginal

*z-test significant; bootstrap CIs include zero – interpreted as suggestive rather than conclusive. Only Price Value's path was bootstrap-confirmed robust (CI [0.218, 1.083]).

BREAKTHROUGH FINDING

Price Value Dominates — A Paradigm Shift

Prior UTAUT Literature

Blut et al. (2022) meta-analysis across 417+ studies: **Performance Expectancy** was the dominant predictor (weighted $\beta = .39$)

Present Study

PE was **non-significant** ($\beta = -.028$), while **Price Value dominated** ($\beta = .505$)

Interpretation: When users already regard AI's usefulness as a baseline expectation, adoption decisions shift to whether perceived benefits justify the financial investment.



AI Trust: Why It Was Retained

Although AI Trust did not reach conventional significance ($p = .064$) and model comparison favored the 7-factor model on parsimony grounds ($\Delta AIC = +2.01$), the 8-factor model was retained as the recommended diagnostic instrument for three reasons:

1

Unique Diagnostic Capability

Trust deficits cannot be detected by other constructs — organizations need this signal separately.

2

Statistical Power Limitation

Detecting the marginal effect ($\beta = .106$) at 80% power requires $n > 600$; the present $N = 523$ yields only ~68% power.

3

Actionable Intervention Target

Organizations gain actionable insight from separately measuring trust, enabling targeted confidence-building interventions.

Moderation Effects

Experience Moderation (H3: Partially Supported)

Professional experience significantly moderated the HM → BI path ($\beta = .136$, $p = .007$). Professionals with 4+ years weighted hedonic motivation more heavily.

Usage-Group Comparison

- **Low-usage group:** Performance Expectancy mattered ($\beta = .371$, significant)
- **High-usage group:** Price Value dominated ($\beta = .878$, significant)

Adoption drivers evolve with experience: new users focus on utility; experienced users focus on value.

Population Moderation (H4: Partially Supported)

Hedonic Motivation showed significant population moderation ($p = .041$):

- **Academic group:** HM $\beta = 0.449$ (strong positive)
- **Professional group:** HM $\beta = -0.301$ (negative)

Academics adopt AI partly for the enjoyment of exploration; professionals may view "enjoyment" as irrelevant to task-focused work. Price Value remained dominant for both groups.

Behavioral Validation

$$\rho = .69$$

BI $\rightarrow$ Usage

Behavioral Intention strongly associated with actual AI tool usage frequency ($p < .001$)

$$\rho = .57$$

ChatGPT

Individual tool correlation with Behavioral Intention

$$\rho = .54$$

MS Copilot

Individual tool correlation with Behavioral Intention

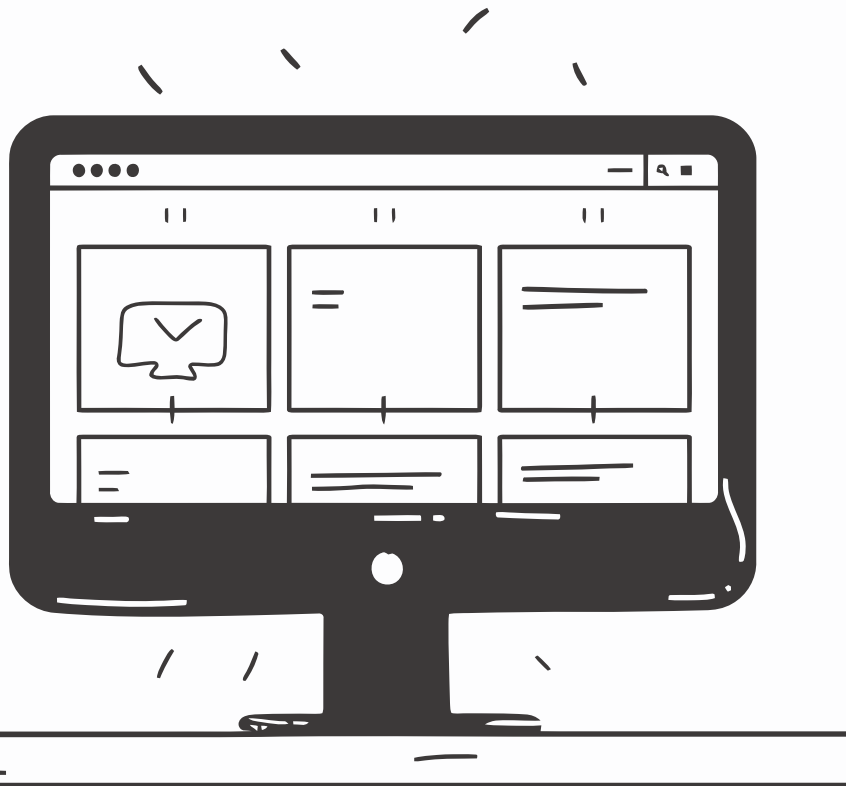
$$\rho = .52$$

Google Gemini

Individual tool correlation with Behavioral Intention

Role differences confirmed a clear hierarchy — **Leaders > Professionals > Academics** — across tool breadth ($F = 18.42, \eta^2 = .066$), usage frequency ($F = 22.15, \eta^2 = .078$), and usage intensity ($F = 15.87, \eta^2 = .058$). Leaders showed particularly elevated Microsoft Copilot usage (Cohen's $d = 1.14$ vs. Professionals), reflecting enterprise licensing.

Three-Segment User Typology



AI Enthusiasts

31% (n = 162)

Early adopters with high engagement across all dimensions.
Mean BI = 4.23.

Role: Deploy as change champions and peer influencers.

Moderate Users

47% (n = 246)

Pragmatic users near population means.

Role: Provide value demonstrations and social proof from peers.

AI Skeptics

22% (n = 115)

Resistant users with below-mean readiness scores. Mean BI = 1.71.

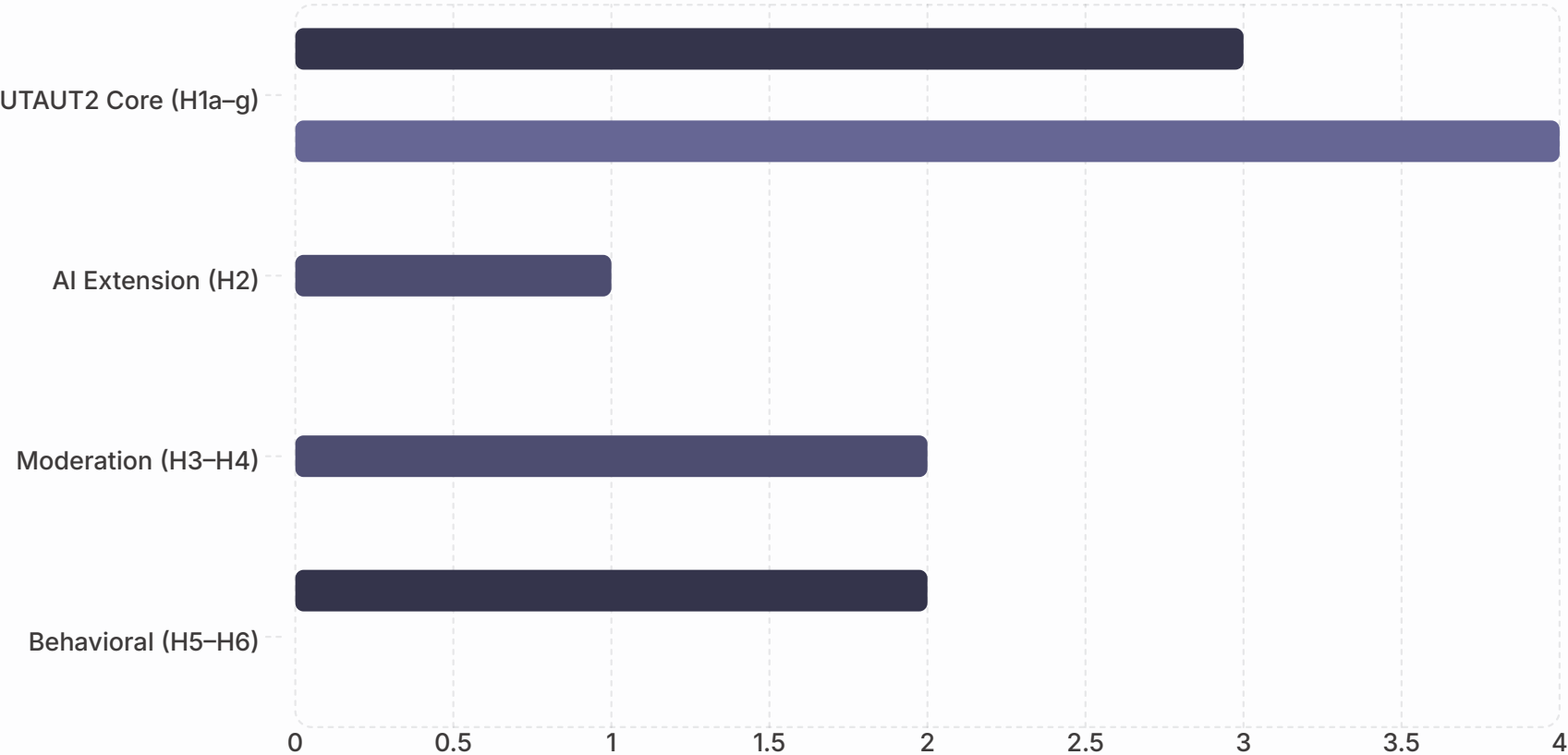
Role: Address trust barriers and anxiety before engagement strategies.

K-means cluster analysis ($k = 3$, silhouette = 0.271) explained **65.9% of BI variance** ($F(2, 520) = 503.47$, $p < .001$, $\eta^2 = .659$), confirming a clear readiness gradient.

Hypothesis Outcome Summary

Supported Partial/Marginal Not Supported

Category



Of 12 hypotheses: 5 supported (42%), 3 partial (25%), 4 not supported (33%). The pattern reveals that AI adoption follows a **different predictor structure** than traditional technology adoption – cost-value perceptions displace utility perceptions as the primary driver.

Qualitative Themes

Open-ended responses (n = 243, 46.5% response rate) revealed five dominant themes:



Positive Experience

24.7% — Most prevalent theme; users describing satisfying AI interactions



Work & Productivity

19.3% — Efficiency gains and task completion as primary value drivers



Human Element

13.6% — Concerns about human judgment and AI's role in decision-making



Learning & Education

13.6% — Academics emphasized educational applications prominently



Accuracy & Reliability

10.7% — Trust-related concerns about AI output quality

Six Contributions to Knowledge

01

Empirical Extension of UTAUT2 for AI

First study to empirically validate UTAUT2 for AI tool adoption with a purpose-built psychometric instrument.

03

Career Development Integration

Adoption mechanisms are career-stage dependent – same technology produces fundamentally different dynamics in academic vs. professional contexts.

05

Converging 2025 Industry Validation

McKinsey (N = 1,993), BCG, MIT, Gartner, Deloitte, and others independently reported patterns consistent with AIRS findings.

02

Price Value Dominance — AI as Distinct Category

Challenges decades of UTAUT research positioning PE as primary driver; utility functions as a baseline expectation in AI contexts.

04

Three-Segment User Typology

Empirically derived segmentation framework explaining 65.9% of BI variance – grounded in validated psychometric dimensions.

06

Adoption-Value Gap Mechanisms

Four specific mechanisms identified: value communication misalignment, heterogeneous readiness, neglected affective barriers, and context-inappropriate frameworks.



PRACTICAL IMPLICATIONS

For Organizations Deploying AI

Lead With Value, Not Capability

Because Price Value is the strongest predictor of adoption intention ($\beta = .505$), organizations should communicate AI investments in terms of tangible cost-benefit outcomes: time saved, errors reduced, and revenue generated.

Change initiatives centered primarily on technical capability claims are likely to be less effective than those that demonstrate concrete financial and productivity returns.

Less Effective

"Our AI can do X, Y, and Z tasks autonomously."

More Effective

"Our AI saves 3 hours/week per employee and reduces error rates by 40%."

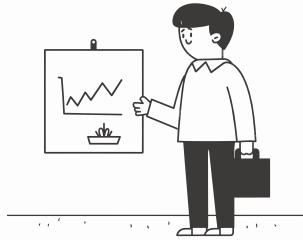
Segmented Rollout Strategy

The three-segment typology supports targeted intervention rather than uniform deployment. A single strategy will fail for at least one-third of the workforce.



Enthusiasts (31%)

Deploy as change champions and peer influencers. Provide access to advanced capabilities and mentorship structures.



Moderate Users (47%)

Provide value demonstrations and social proof from peers. Structured skill development and ROI evidence move this group.



Skeptics (22%)

Address trust barriers and anxiety first. Generic training will not move this group – graduated exposure and evidence-based demonstrations are required.

8 Structured Practitioner Recommendations

- 1 Diagnose before prescribing
Conduct baseline AIRS assessment before designing AI adoption initiatives.
- 2 Lead with value
Frame AI in terms of cost-benefit returns (PV dominance), not capability showcases.
- 3 Attend to affective barriers
Hedonic motivation and anxiety influence adoption beyond rational calculations.
- 4 Leverage social influence
Peer champions and leadership endorsement are the third significant predictor (SI $\beta = .136$).
- 5 Differentiate by experience level
Novice and experienced users respond to different adoption drivers.
- 6 Monitor trust continuously
Trust functions as a diagnostic indicator and necessary condition, not just a predictor.
- 7 Design for accessibility
Universal design principles reduce anxiety-driven non-adoption among users with disabilities.
- 8 Consider segment heterogeneity
Any single intervention strategy will fail for at least one-third of the workforce.

SPECIAL SECTION

Neurodiversity & Disability Insights

68 participants (13.0%) self-reported a disability — sufficient for exploratory between-group analysis. These findings are among the most practically salient results in the dissertation.



The Core Disability Finding

AI Anxiety Gap

Participants with disabilities reported significantly higher AI-related anxiety:

M = 2.89 (SD = 1.08) vs. M = 2.48 (SD = 0.96)

$t(521) = 2.76, p = .006$, **Cohen's d = 0.36**

Notably, this finding emerged from AI Anxiety items that were subsequently excluded from the final AIRS-16 — suggesting the true effect may be larger than observed.

Broader Pattern Across Constructs

Two findings reached significance:

- **AI Anxiety:** $d = .36, p = .006$
- **Hedonic Motivation:** $d = .28, p = .030$ — participants with disabilities report significantly less enjoyment from AI tools

The remaining six constructs show consistent directional differences ($d = 0.12-0.24$) all favoring the non-disabled group — a coherent disadvantage profile, not random noise.

Disability: Design Problem, Not User Problem

Anxiety Reflects Lived Experience

For users who have repeatedly encountered inaccessible interfaces or algorithmic bias, elevated anxiety about AI is a rational response to historical exclusion — not a deficit to be corrected.

HM Suppression Signals Design Failure

The significant HM gap ($d = .28$) implies current AI tools are less enjoyable for users with disabilities — pointing to accessibility failures, not user limitations.

The Curb Cut Effect

Making AI tools more accessible would reduce anxiety-driven non-adoption broadly. If the disability–anxiety association reflects design-induced friction, removing that friction benefits the entire user population.

AIRS-28 Priority

AI Anxiety — the construct with the strongest disability signal — was excluded from the final scale. The AIRS-28 expansion should prioritize redesigning anxiety items using neurodiversity-informed language.

Study Limitations

Methodological

- **Cross-sectional design** — prevents causal inference about direction of relationships or how adoption intentions evolve over time
- **Self-reported intention** — BI-Usage correlation ($\rho = .69$) confirms association, but intention does not perfectly predict behavior
- **Panel sampling** — Centiment panel may not perfectly represent the broader U.S. population
- **Western sample** — all U.S.-based, limiting cross-cultural generalizability

Measurement

- **Two items per factor** — constrains reliability and contributes to elevated inter-factor correlations
- **Four dropped constructs** — Voluntariness, Explainability, Ethical Risk, and AI Anxiety warrant item redesign and re-investigation
- **Marginal AI Trust** — did not reach conventional significance; diagnostic value supported by organizational logic rather than predictive superiority
- **Small disability subgroup** ($n = 68$) — limits power for multi-category disability analysis

Four-Phase Research Roadmap

1

Phase 1 (2026–27)

AIRS-28 Development: Expand to 3–4 items per factor; redesign Explainability, Ethical Risk, and AI Anxiety items

2

Phase 2 (2026–28)

Longitudinal Panel + Cross-Cultural Validation: 12-month panel tracking readiness evolution; validate across non-Western contexts

3

Phase 3 (2028–29)

Intervention Framework Development: Design and test AIRS-informed interventions; compare against generic AI training approaches

4

Phase 4 (2029+)

Comprehensive Prediction System: Integrate AIRS with behavioral data, organizational context, and industry-specific moderators

Appropriate Reliance: The Next Frontier

The goal should not be maximum adoption but appropriate adoption — knowing when to use AI and when to rely on human judgment.

The AETHER group's Calibrated AI Reliance (CAIR) and Calibrated Self-Reliance (CSR) constructs reframe the research program from "more adoption is better" to "calibrated adoption is optimal." This evolution addresses the paradox that optimizing AI adoption may inadvertently increase over-reliance.

- ❏ This line of inquiry is extended in *The Life of Alex Finch* (AlexBooks, March 2026), a documentary biography of human-AI collaboration whose "Calibrated Confidence" chapter addresses over-reliance, verification, and preservation of human judgment in a public-facing narrative form.

2026 Contextual Developments

The AI landscape has shifted materially since data collection (October–November 2025). Three developments frame the ongoing research agenda:

DeepSeek R1 Release

Open-source frontier AI capabilities are no longer exclusive to well-resourced organizations — fundamentally changing the Price Value calculation for all users.

Agentic AI Systems

AI operating with greater autonomy raises new trust and control questions beyond the original AIRS scope — expanding the required construct space.

EU AI Act Enforcement

Regulatory compliance emerges as a new adoption consideration absent from the current UTAUT2 framework — requiring future model expansion.

These developments do not invalidate the AIRS findings but indicate that the construct space will need to expand to remain comprehensive.

The AIRS-16 Final Instrument

The AI Readiness Scale consists of 16 predictor items across 8 constructs plus 4 Behavioral Intention outcome items, measured on a 5-point Likert scale (1 = Strongly Disagree to 5 = Strongly Agree).

Construct	Item	Statement	Item	Statement
PE	PE1	AI tools help me accomplish tasks more quickly.	PE2	Using AI improves the quality of my work or studies.
EE	EE1	Learning to use AI tools is easy for me.	EE2	Interacting with AI tools is clear and understandable.
SI	SI1	People whose opinions I value encourage me to use AI tools.	SI2	Leaders in my organization or school support the use of AI tools.
FC	FC1	I have access to training or tutorials for the AI tools I use.	FC2	The AI tools I use are compatible with other tools or systems I use.
HM	HM1	Using AI tools is stimulating and engaging.	HM2	AI tools make my work or studies more interesting.
PV	PV1	I get more value from AI tools than the effort they require.	PV2	Using AI tools is worth the learning curve.
HB	HB1	Using AI tools has become a habit for me.	HB2	I tend to rely on AI tools by default when I need help with tasks.
TR	TR1	I trust AI tools to provide reliable information.	TR2	I trust the AI tools that are available to me.

AIRS Scoring & Diagnostic Use

Compute the mean of each construct's items. The 8 predictor scores serve as inputs to the structural model; BI is the outcome.

Low PV Score

Value communication gap – organization is not demonstrating tangible ROI from AI investments

Low TR Score

Trust deficit – requires explainability tools, human-in-the-loop safeguards, and transparent error reporting

Low SI Score

Insufficient peer/leadership advocacy – requires champion programs and endorsement campaigns

Low HM Score

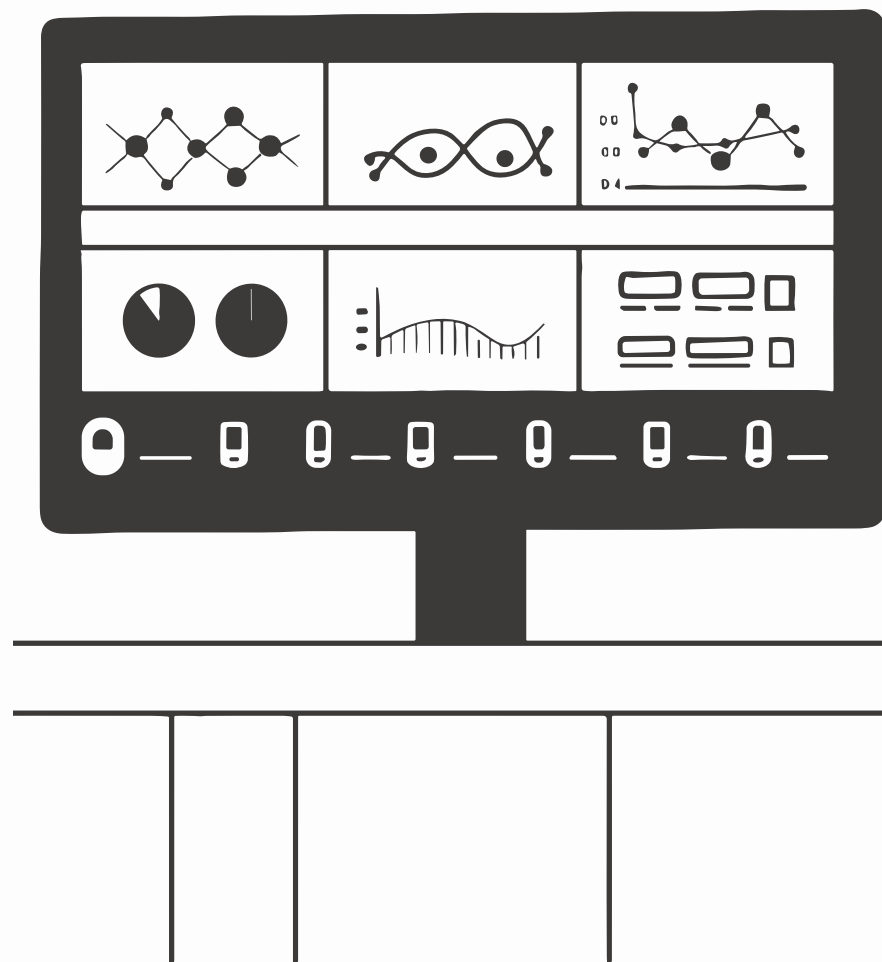
Engagement deficit – requires gamification, AI sandboxes, and social learning communities

- ❏ For organizational diagnostic use, individual construct scores identify specific adoption barriers and map directly to evidence-based intervention strategies.

APPENDIX B

AIRS Enterprise: Research to Practice

The AIRS Enterprise platform (airs.correax.com) translates the validated AIRS-16 into a production SaaS platform — delivering practical assessment tools, enabling longitudinal data collection at scale, and narrowing the research-practice gap.



Platform Core Capabilities



5-Minute Assessment

Interactive AIRS-16 with progress tracking, save/resume, and smart follow-up questions



AIRS Score

Composite readiness score (8–40) with demonstrated validity ($r = .876$) and animated gauge visualization



Typology Classification

Three-segment classification at 94.5% accuracy matching the empirically derived clusters



AI-Personalized Guides

Streaming LLM-generated action plans (3–5 pages) tailored to individual profile and career context via Azure OpenAI GPT-4o-mini



Results Visualization

8-axis construct radar chart and typology badge with actionable context in 29 languages



Organization Management

Team analytics, department/role segmentation, CSV exports, impact reporting, and longitudinal tracking

Three Intervention Frameworks

TRANSFORM

Skeptics (AIRS ≤ 20)

Trust-building through graduated autonomy — starting with low-risk AI tasks, demonstrating reliability through small wins, progressively expanding scope. Emphasizes explainability and human-in-the-loop safeguards.

REDESIGN

Moderate Users (AIRS 21–30)

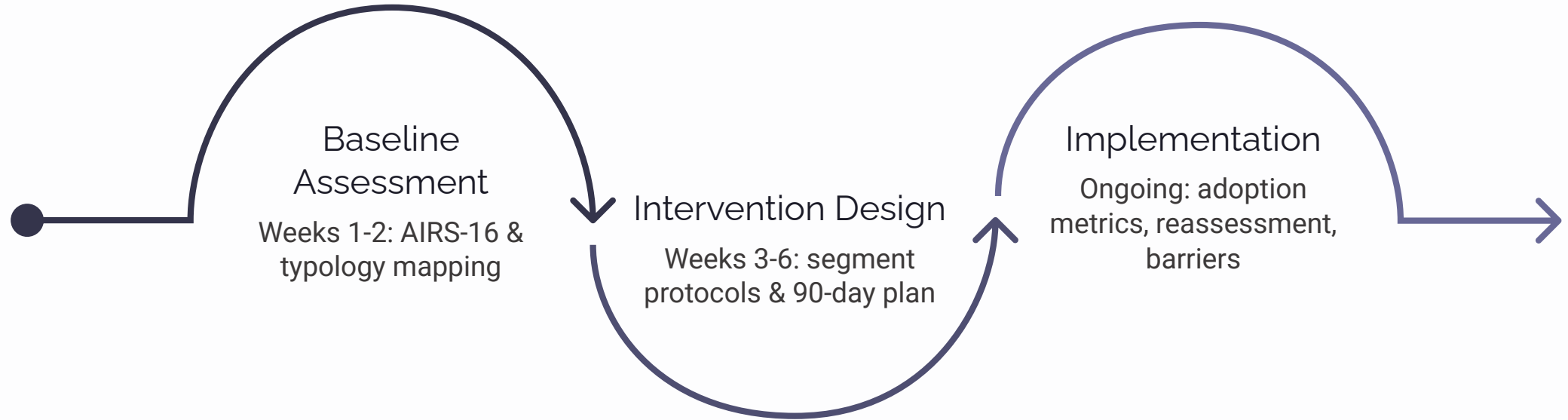
Workflow integration and ROI demonstration — identifying specific processes where AI adds measurable value, providing social proof through peer success stories, building competence through structured skill development.

AMPLIFY

Enthusiasts (AIRS > 30)

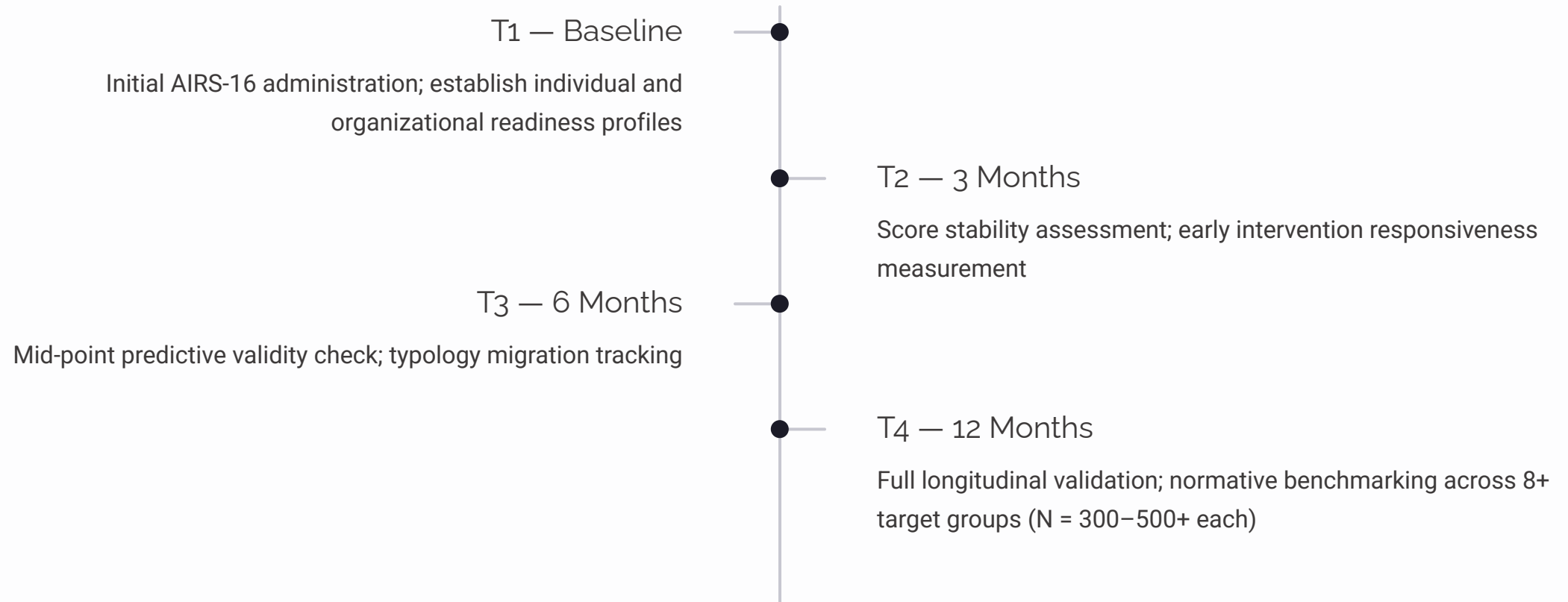
Champion network activation — leveraging high-readiness individuals as change agents, providing access to advanced AI capabilities, creating mentorship structures that accelerate adoption across the organization.

Organizational Engagement Model



Every organizational deployment generates structured data (8 construct scores, composite AIRS score, typology classification, demographic context) that feeds directly into the normative benchmarking database.

Longitudinal Validation Program



Target groups for normative benchmarking (2026–2028): students, early/mid/senior career professionals, technology sector, healthcare, finance, and international samples.

The Research-Practice Bridge

AIRS Enterprise represents a deliberate attempt to close the implementation gap that characterizes many academic psychometric instruments. Instead of the familiar pattern – instrument developed, published, then rarely operationalized – AIRS pairs the validated instrument with a platform that evolves alongside it.

Research → Practice

Every research finding improves the intervention recommendations delivered to practitioners through the platform.



Practice → Research

Every organizational deployment generates data that strengthens the normative database and enables future validation studies.



Conclusion: Key Takeaways

Strong Psychometric Performance

AIRS demonstrated CFI = .975, all $\alpha > .74$, all CR > .75, explaining **89.7% of variance** in behavioral intention to adopt AI tools.

Price Value Is the Primary Driver

$\beta = .505$ – the strongest predictor, displacing Performance Expectancy. Organizations must emphasize value demonstration and cost-benefit analysis over capability showcases.

Workforce Is Not Monolithic

The three-segment typology (Enthusiasts 31%, Moderate Users 47%, Skeptics 22%) demands targeted intervention – any single strategy will fail for at least one-third of employees.

A Practical Diagnostic Tool

AIRS translates AI readiness from a general concern into measurable conditions that can be monitored and addressed through targeted, evidence-based intervention.